InterviewIQ

AI-Powered Multimodal Mock Interview Evaluation Platform

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01 ABSTRACT

InterviewIQ is a web-based multimodal mock-interview coaching platform that evaluates technical answer correctness, vocal emotion, and temporal visual behavior.

The system separates Technical Score, Model Classification Confidence, and behavioral delivery indicators to provide interpretable feedback without treating classifiers as abilities, probabilities as direct measurements of competence, or personal confidence.

NLP + AUDIO + VISION

02 AIM OF THE WORK

>> **Develop an interpretable AI-powered** multimodal mock-interview platform

>> **Provide structured feedback on Technical** Answer Correctness, Vocal Delivery, and Visual Behavioral Patterns.

>> **Position the system as an Interview** Coaching & Research Prototype, not an Autonomous Hiring System.

03 DATASET

BAVED – AUDIO

(mic)

- 1,935 Audio Recordings
- 60 Speaker IDs
- 7 Arabic Words

Split	Actors	Samples
Train	1–18	3,656
Valid.	19–21	624
Test	22–24	624

RAVDESS – VISUAL

(vid)

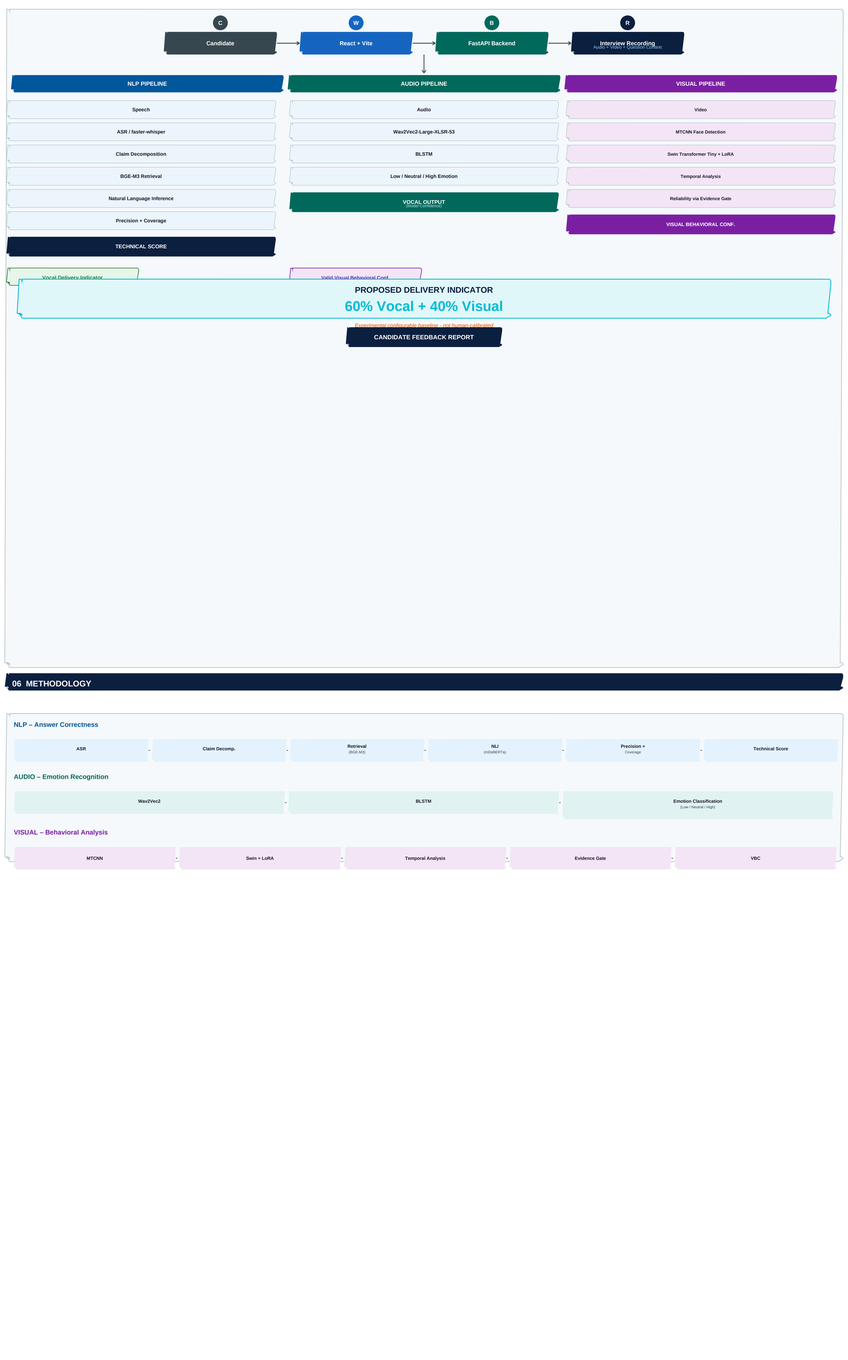
- 24 Professional Actors
- 8 Acted Emotion Categories

Split	Actors	Samples
Train	1–18	3,656
Valid.	19–21	624
Test	22–24	624

04 USED TOOLS & TECHNOLOGIES

React	FastAPI	Python	SQLAlchemy
Whisper	BGE-M3	(Embedding)	mDeBERTa
Wav2Vec2	BLSTM	PyTorch	TCN
Swin	Transformer	LoRA / PEFT	

05 SYSTEM OVERVIEW



06 METHODOLOGY

NLP – Answer Correctness

ASR → Claim Decomp. → Retrieval (BGE-M3) → NLI (mDeBERTa) → Precision + Coverage → Technical Score

AUDIO – Emotion Recognition

Wav2Vec2 → BLSTM → Emotion Classification (Low / Neutral / High)

VISUAL – Behavioral Analysis

MTCNN → Swin + LoRA → Temporal Analysis → Evidence Gate → VBC

07 RESULTS

AUDIO – BAVED N809 (Source-Domain Results)

89.15%	89.19%	86.36%
TEST ACCURACY	MACRO-F1	NEUTRAL PRECISION

73.88%	73.24%
VALIDATION ACCURACY	TEST ACCURACY
72.29%	68.85%
WEIGHTED F1	MACRO-F1

NLP – REAL CLAIM DISTRIBUTION MISMATCH

Gold Set Macro-F1

Zero-shot	LoRA Adapter
0.880	0.850

Real-Claims Coverage

Zero-shot	LoRA Adapter
0.395	0.180

08 PLATFORM ACCESS

09 QR & DEMO ACCESS

Presentation / Demo

GitHub Repository

10 CONCLUSION

>> **InterviewIQ demonstrates the feasibility of** an interpretable multimodal mock-interview coaching architecture.

>> **Integrates NLP, Audio, Vision, and Web** technologies successfully.

>> **Separates Technical Correctness from Delivery** Indicators and preserves interpretability.

>> **Current experimental results support the** research prototype but do not yet establish unrestricted real-world hiring validity.

>> **Provides a credible engineering foundation for** explainable AI-assisted interview coaching.

11 FUTURE WORK

- Real-time interview analysis
- Multi-language support
- HR / ATS integration
- Advanced transformer-based emotion detection